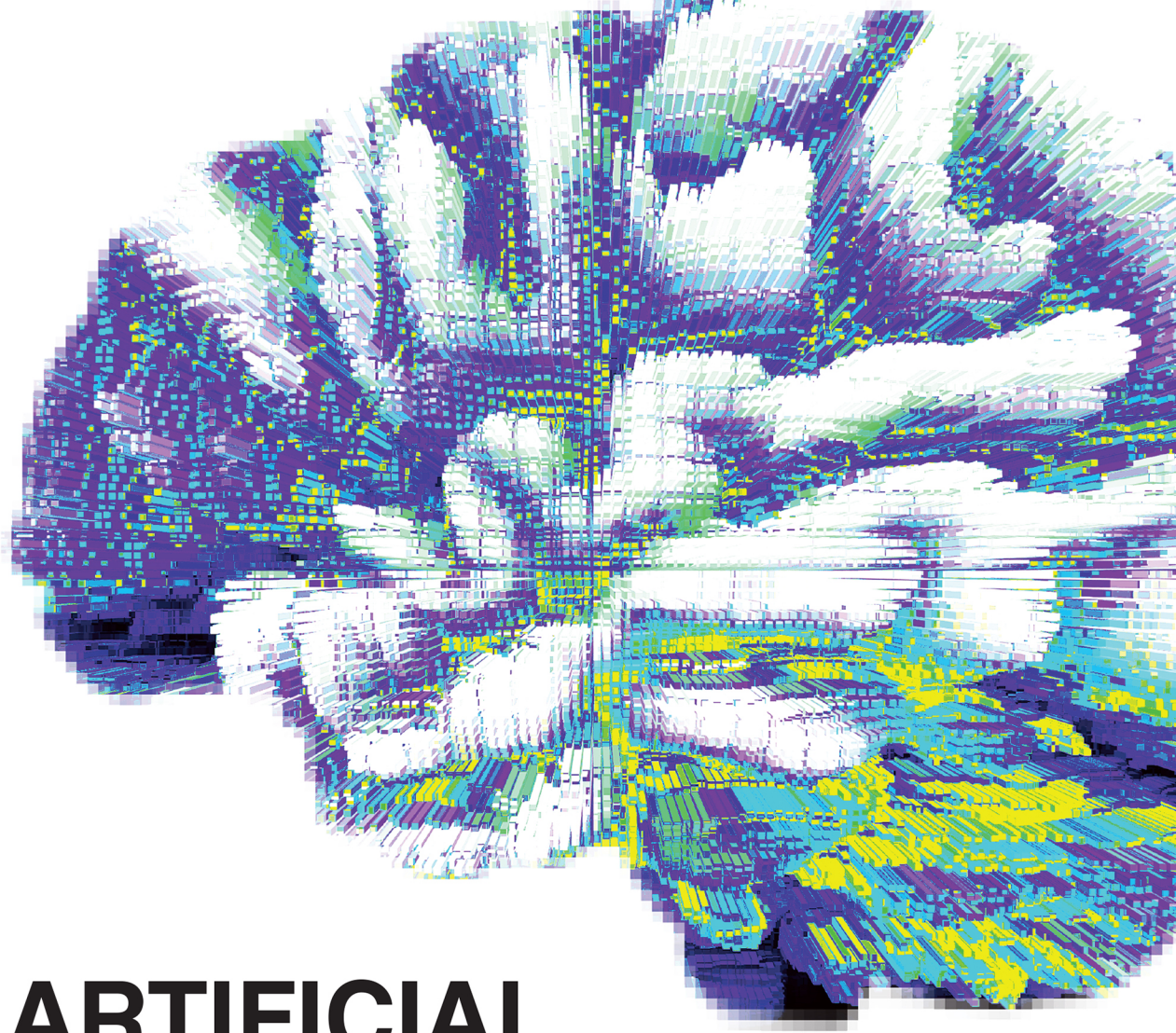




ARTIFICIAL INTELLIGENCE

IN PERFORMANCE-DRIVEN DESIGN

THEORIES, METHODS, AND TOOLS

Edited by

NARJES ABBASABADI • MEHDI ASHAYERI

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Artificial Intelligence in Performance-Driven Design

Artificial Intelligence in Performance-Driven Design

Theories, Methods, and Tools

Edited by

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Contents

List of Contributors *xi*

Introduction *xiii*

1 Augmented Computational Design *1*

Introduction *1*

Background *2*

Relevance of AI in AEC *2*

Historical Context *3*

Design as Decision-Making *5*

AI for Generative Design *7*

Framework *9*

Design Space Exploration *11*

Spatial Design Variables *13*

Statistical Approaches to Design *14*

Demonstration *15*

Case Study *15*

Methodology *16*

Results *21*

BBN Validation Results *21*

Toy Problem *22*

Discussion *22*

Outlook *25*

Acronyms *26*

Notations *27*

References *28*

2 Machine Learning in Urban Building Energy Modeling *31*

Introduction *31*

Urban Building Energy Modeling Methods *32*

Top-Down Models *33*

Bottom-Up Models *33*

Uncertainty in Urban Building Energy Modeling *36*

Epistemic Uncertainty *36*

Stochastic Uncertainty *36*

Addressing Uncertainty *37*

Machine Learning in Urban Building Energy Modeling *39*

Supervised Learning	39
Unsupervised Learning	44
Reinforcement Learning	46
Machine Learning-Based Surrogate UBEM	47
Conclusion	49
References	50
3 A Hybrid Physics-Based Machine Learning Approach for Integrated Energy and Exposure Modeling	57
Introduction	57
Materials and Methods	59
Data, Data Sources, and Dataset Processing	59
Methodology	61
Results	70
Physics-Based Simulation	70
<i>Data-Driven Computation (Prediction)</i>	70
Discussion	73
Conclusion	74
Acknowledgment	75
References	75
4 An Integrative Deep Performance Framework for Daylight Prediction in Early Design Ideation	81
Introduction	81
Background	83
Daylight Simulation	84
Deep Learning Models	85
DL-Based Surrogate Modeling	85
Verification Methods	85
Research Methods	86
Data Acquisition	86
Model Training	88
Results and Validation	88
Discussions of Results	90
Conclusions	94
References	94
5 Artificial Intelligence in Building Enclosure Performance Optimization: Frameworks, Methods, and Tools	97
Building Envelope and Performance	97
Artificial Intelligence and Building Envelope Overview	97
Optimization Routes and Building Envelope	98
Optimization Frameworks	99
Optimization Methods	99
Machine Learning and Building Envelope	101
Artificial Neural Network	101
Convolutional Neural Network	105

	Recurrent Neural Network	105
	Generative Adversarial Networks	106
	Ensemble Learning	107
	Discussions on Practical Implications	108
	Summary and Conclusion	109
	References	110
6	Efficient Parametric Design-Space Exploration with Reinforcement Learning-Based Recommenders	113
	Introduction	113
	Methodology	115
	Section 01: Clustering Design Options	116
	Section 02: Reinforcement Learning-Based Recommender System	120
	Design Dashboard	123
	Discussion	124
	Conclusion	125
	References	126
7	Multi-Level Optimization of UHP-FRC Sandwich Panels for Building Façade Systems	129
	Introduction	129
	Building Façade Design Optimization	130
	Methodology	134
	Midspan Displacements and Thermal Resistivity of UHP-FRC Panels	136
	Energy Performance of the UHP-FRC Panels at the Building Level	141
	Life Cycle Cost Analysis of the UHP-FRC Panels	142
	Surrogate Models	145
	Multi-objective Optimization Algorithm	147
	Results and Discussion	148
	Surrogate Models	148
	Pareto Front Solutions	151
	Conclusion	152
	References	153
8	Decoding Global Indoor Health Perception on Social Media Through NLP and Transformer Deep Learning	159
	Introduction	159
	Literature Review	161
	Social Media and Urban Life: Theories, Challenges, and Opportunities	161
	Methods for Computing Social Media Data in Environmental Studies	163
	Materials and Methods	168
	Data Query	168
	Text Preprocessing	169
	Text Tokenization	169
	Text Summarization	170
	Generating Co-occurrence Matrix	170
	Sentiment Analysis and Classification	170

Visualizations	171
Embedding Visualization	171
Attention Score Visualization (Attention Map) and Interpretation	172
Results and Discussion	173
Conclusion	178
References	179
9 Occupant-Driven Urban Building Energy Efficiency via Ambient Intelligence	187
Introduction	187
Occupancy and Building Energy Use	191
Definitions	191
Occupant Monitoring Methods	193
Occupant Monitoring Via Observational Studies	194
Occupant Monitoring via Experimental Studies	195
Occupant-driven Energy Efficiency via Ambient Intelligence	196
Ambient Intelligence	196
AmI-Based Energy Efficiency Feedback (EEF) Systems	197
Energy Efficiency via AmI Systems and Digital Twins Technology	201
Conclusion	202
References	203
10 Understanding Social Dynamics in Urban Building and Transportation Energy Behavior	211
Introduction	211
Methodology	213
Modeling Framework	214
Explanatory Model	214
Data	215
Results and Discussion	219
Effects of Occupancy and Socio-economic Factors	219
Variable Importance (VI)	219
Lek's Profile	219
Conclusion	226
References	227
11 Building Better Spaces: Using Virtual Reality to Improve Building Performance	231
Introduction	231
Applications of Virtual Reality in Building Performance	233
Virtual Reality for Improving Building Design through Integrated Performance Data	233
Virtual Reality for Building Design Reviews and Education in Architecture and Engineering	236
Virtual Reality for Research on Building Occupant Comfort and Well-Being	240
Conclusion	243
References	245

12 Digital Twin for Citywide Energy Modeling and Management	251
Introduction	251
Urban Building Energy Digital Twins (UBEDTs)	252
Definition and Conceptualization	252
Implications for Citywide Energy Management	254
Enabling Technologies	256
Twining Technologies	256
Urban Digital Twin (UDT) and Data Sources	258
Artificial Intelligence (AI) and Digital Twin	260
Relationship Between IoT, Big Data, AI–ML, and Digital Twins	261
Interoperability Technologies	262
Maturity Levels	263
Architecture	265
Data Acquisition Layer	266
Transmission Layer	266
Modeling and Simulation Layer	266
Data/Model Integration Layer	269
Service/Actuation Layer	269
Challenges in Implementing Citywide Digital Twins	269
Data Quality and Availability	270
Required Smart Infrastructure and Associated Cost	270
Interoperability	270
Data Analysis	271
Cybersecurity and Privacy Concerns	271
Conclusion	272
References	272
Index	277

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Introduction

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Artificial intelligence (AI) has emerged as a ubiquitous force in today's world, asserting its influence across an expansive spectrum of human endeavors. AI, as defined by Margaret A. Boden in *AI: Its Nature and Future*, "seeks to make computers do the sorts of things that minds can do" (Boden 2016). At its core, AI endeavors to engineer computational entities capable of performing tasks traditionally associated with human cognition and to facilitate a synthetic replication of human abilities such as communication, learning, perception, problem-solving, and reasoning. In pursuit of this, the discipline is methodically segmented into distinct yet interconnected subfields, for example: machine learning (ML), which learns from data to enable informed decision-making; natural language processing (NLP), which deciphers and constructs human language; computer vision (CV), which interprets visual data; and robotics, which imparts autonomy to machines for task execution. These diverse subfields underpin AI's quest to forge machines that operate with a semblance of human intelligence, enhancing human capacities and spearheading a revolution in technological innovation.

The history of AI unfolds from Blaise Pascal's 17th-century mechanical calculator through the 19th-century programmable machine of Charles Babbage and Ada Lovelace, arriving at the 20th century where the formalization of neural networks by Warren McCulloch and Walter Pitts, and Alan Turing's machine intelligence, sets the stage for AI's theoretical and practical evolution. Alan Turing's seminal work, "Computing Machinery and Intelligence," is central to AI's philosophical underpinnings, which posits the critical inquiry, "Can machines think?" and introduces the Turing Test as a measure of a machine's capability to exhibit intelligence indistinguishable from human intelligence (Turing 1950). Turing's work has significantly influenced the computational cognition and the trajectory of AI evolution, underscoring Turing's enduring legacy in shaping the ongoing discourse in computing technology.

The term "Artificial Intelligence" was coined in 1956, around the time of the seminal Dartmouth Conference, and has since become the standard bearer for a field characterized by rapid growth. It has navigated through the vicissitudes of "AI winters" and resurgences, marked by significant milestones. This journey includes the triumphs of chess-playing computers such as IBM's Deep Blue, the widespread adoption of digital assistants such as Siri, Alexa, and Google Assistant, and the development of advanced generative models in the 21st

century, particularly those in the realm of deep learning enabling the generation of text, images, and other forms of media to open up new possibilities for creativity, automation, and problem-solving. Moreover, the integration of AI with big data has catalyzed significant breakthroughs across various domains, with ML and deep learning technologies becoming intertwined with the ability to process and learn from vast datasets and enable complex predictive models. In addition, it led to significant advances in autonomous systems, including self-driving cars, drones, and robotics, as well as in NLP technologies like GPT, and biologically impactful models such as DeepMind's AlphaFold. Today, as AI navigates ethical landscapes and quantum futures, it stands not only as a testament to the relentless human pursuit of creating machines that mirror and extend our cognitive faculties but also as a beacon for the untapped capabilities of intelligent systems and continuously expanding the horizons of computational and cognitive potential.

While data-driven approaches using AI have shown their power to solve challenges across many domains, particularly science and engineering, their full potential within the built environment, specifically architectural discipline, remains largely underexplored. AI innovations and advancements in sensing technologies, real-time global information exchange, and open data initiatives have influenced our cities and increased our ability to better understand and solve interconnected built environment problems. These advancements, combined with predictive information modeling and intelligence across scales, present opportunities to address multi-scale crucial issues—from mobility and energy to environmental quality and human well-being—potentially enabling cities to be transformative stages. The integration of AI and big data into urban and building modeling workflows facilitates enhancing the accuracy, efficiency, scalability, and accessibility of complex models and enabling a deeper understanding and more nuanced solutions to multifaceted problems. Particularly, the convergence of AI and big data with built environment technologies presents a transformative opportunity for performance-driven design and planning, promising more accurate, faster, and actionable insights that could lead to improved decision-making. In this light, AI emerges as a catalyst for reimagining the future of built environments, driving performance-driven design toward greater social and environmental sustainability.

This book, *Artificial Intelligence in Performance-Driven Design: Theories, Methods, and Tools*, serves as an interdisciplinary exploration of the utilization of AI, specifically ML, in the domain of performance-driven design and modeling within the context of the built environment. This work delves into the theoretical foundations and methodological frameworks that underpin the application of AI techniques in performance-driven exploration. Notably, it places a particular emphasis on the multi-scale modeling of diverse facets such as energy flows, environmental quality, and human systems. This book examines relevant examples showcasing the integration of AI and big data into modeling and design workflows to enhance their accuracy, efficiency, scalability, and accessibility in complex models. Moreover, it explores conceptual frameworks and computational tools that harness AI's potential in physics-based simulation frameworks, facilitating the integration of physics-based modeling, optimization, and automation procedures, showcasing how AI can streamline decision-making processes in performance-driven design.

Across the various chapters in this book, ML and its application to the built environment offers an array of methodologies tailored to distinct learning paradigms, each leveraging data to enhance design, planning, and operational efficiencies. Fundamentally, ML is categorized

into several core approaches: supervised learning, which employs labeled datasets to instruct algorithms on pattern recognition and outcome prediction, is essential for tasks such as predicting building energy consumption and classifying buildings by energy efficiency levels. This approach has become fundamental for developing predictive models that inform sustainable architectural practices. Conversely, unsupervised learning leverages unlabeled data to discover latent patterns and correlations, enabling applications like the clustering of buildings according to energy usage or the application of dimensionality reduction techniques to distill complex data into actionable insights. Reinforcement learning broadens this spectrum by focusing on the iterative process of decision-making through direct interaction with environments, proving pivotal in scenarios such as the optimization of building façades, building and urban layouts, traffic management, and the operational dynamics of smart grids. Additionally, generative algorithms introduce the capability to synthesize new data samples based on predefined parameters, expanding the creative scope within performance-driven design and planning. Together, these applications underscore the transformative potential of AI and ML in redefining the built environment, pushing the boundaries of traditional architectural paradigms toward more efficient, responsive, and sustainable outcomes.

Deep learning, as a branch of ML and AI, stands at the forefront of technological advancements, transforming the built environment sector. Categorized by neural networks designed to mimic the complex processing layers of the human brain, deep learning surpasses traditional data analysis techniques by its capacity to learn from and interpret vast amounts of data autonomously. This capability enables the identification of complex patterns and relationships within data, significantly enhancing accuracy in predictive analytics, image processing, and autonomous system controls. Deep learning's advanced analytical prowess is particularly beneficial in predictive modeling for various performance metrics, such as energy consumption, and in enhancing automated and optimized design processes. By employing deep learning, designers, planners, and researchers in the built environment are equipped with powerful tools for driving innovation, improving sustainability, and responding to complex design and planning challenges. The integration of deep learning signifies a major advancement in utilizing data for creating smarter, more efficient, and sustainable solutions within the built environment. It also highlights the broad potential of AI and ML technologies to transform and improve how we develop built environments.

The book also explores the integration of intelligent systems and digital twin technology throughout the lifecycle of the built environment. It underscores its transformative potential in enhancing our comprehension and management of built environments. Intelligent systems, augmented with AI capabilities, contribute to real-time monitoring, analysis, and adaptation of building systems to optimize performance. Digital twins, virtual replicas of physical structures and systems, serve as powerful tools for monitoring prediction, and decision support. The integration of AI within these paradigms augments their utility, leading to more informed design and operational decisions that can enhance the sustainability, efficiency, and user experience of the built environment. Through the integration of intelligent systems and digital twins, AI emerges as a catalyst for enhancing our understanding and management of the complexities inherent in contemporary built environments.

Presented herein is a synopsis, capturing the essence and intellectual journey offered by each chapter:

Chapter 1 presents methodological reflections on the necessity and utility of AI in generative design. It also offers a concise history of significant AI developments, establishing a groundwork for understanding architectural design as a series of decision-making processes. This chapter sets the stage for perceiving architectural design as a sequence of decision-making steps. It explores how AI can augment generative design by addressing numerous decisions for achieving specific outcomes or performance indicators. The essence of performance-based generative design lies in creating statistical or simulation-based links between these decisions and their consequences for mapping and navigating through intricate decision landscapes. The chapter further investigates the potential of AI in refining decision-making in architectural design, particularly in mapping and navigating complex design environments.

Chapter 2 explores the integration of ML into urban building energy modeling (UBEM), which is key for quantifying and understanding city-wide energy dynamics. It underscores ML's role in enhancing UBEM's predictive and optimization capabilities, focusing on improved accuracy and efficiency. The chapter reviews current UBEM methods, their strengths, limitations, and inherent uncertainties, showing how ML helps overcome these challenges. It explores various ML paradigms—supervised, unsupervised, and reinforcement learning—and their specific algorithms, as well as ML-based surrogate modeling for more accurate, efficient, and uncertainty-reduced urban energy consumption estimates. This integration aims to advance urban energy modeling facilitating more efficient, informed decision-making in urban development.

Chapter 3 introduces a hybrid approach, combining ML and urban big data analytics with physics-based performance simulation, for the integrated modeling of indoor air quality, operational energy, and ambient airflow in urban buildings. It aims to optimize design by focusing on both climate and health considerations. A pilot was conducted on a large office building prototype in a densely populated urban setting. This approach utilized advanced simulation tools, including computational fluid dynamics (CFD), energy modeling, and artificial neural networks (ANNs). The results demonstrated the framework's effectiveness in evaluating building performance strategies and understanding natural ventilation and indoor air quality. Compared to traditional methods, this hybrid model significantly enhances simulation speed and accuracy. It offers valuable insights for architects and engineers in the early design stages, enabling the integration of occupant health with energy efficiency, contributing to decarbonization efforts.

Chapter 4 investigates the fusion of AI with performance-driven design strategies, focusing on developing an automated, accurate daylight performance simulation method. As part of broader efforts to automate environmental assessments in design, it introduces "Deep Performance" (DP), a deep learning-based method for creating a surrogate model. This model, trained on vast datasets of building plans and daylight meshes, predicts architectural features and daylight distribution. The DP method showcases notable efficiency, being 600 times faster than traditional simulations, and achieves 90% accuracy. The shift from simulation-based to prediction-based approaches is transformative, enabling designers to integrate real-time environmental metrics into generative design systems. Early incorporation of daylight analysis in design decisions aids in exploring passive strategies and optimizing layouts and envelope designs. Ultimately, this research aims to empower environmentally

efficient and cost-effective designs in the built environment through advanced, data-driven solutions.

Chapter 5 reviews the role of AI in optimizing building enclosure performance, a critical aspect influencing a building's overall efficiency. Building enclosures significantly affect thermal, daylighting, acoustics, and indoor air quality performances, yet their complex design and varying environmental factors make optimal solutions challenging. AI automates the design and evaluation process, enhancing information synthesis and decision-making. As a result, architecture has increasingly incorporated AI and computational optimization into building enclosure design. It provides an overview of AI frameworks, methods, and tools, with a focus on ML algorithms, especially neural networks, and optimization methods like metaheuristic algorithms. These are specifically discussed in the context of designing and analyzing building envelope performance, showcasing their potential for improving sustainable building practices.

Chapter 6 introduces a reinforcement learning-based recommender for efficient exploration in parametric design. Parametric designs, while flexible, often overwhelm designers with numerous alternatives. Traditional optimizers such as genetic algorithms focus mainly on quantitative outcomes, neglecting aesthetic preferences. The proposed framework integrates a reinforcement learning recommender to navigate these vast design spaces effectively. A case study on a mid-rise apartment building demonstrates this. This approach learns from user interactions, suggesting designs that align with both performance and aesthetic choices. It streamlines the decision-making process, enabling designers to explore high-performing solutions with personal design preferences within a vast array of options.

Chapter 7 explores multi-level optimization of ultra-high-performance fiber-reinforced concrete (UHP-FRC) sandwich panels for building façades using AI, optimizing energy efficiency, structural performance, and construction cost. It analyzes design optimization at building and assembly levels, focusing on façade system variables for building performance and panel-specific criteria. A surrogate-based optimization framework is proposed, providing Pareto efficient solutions across panel displacement, building energy use, and life cycle cost. Illustrated with a case study of an office building, the study employs a genetic algorithm for optimization. The results underscore the importance of holistic design considerations, preventing suboptimal performance in isolated areas. The methodology offers insights into optimizing multi-layer façade systems from various perspectives.

Chapter 8 delves into decoding global indoor health perceptions on social media, utilizing NLP and transformer deep learning techniques like the RoBERTa algorithm. It examines digital media's impact in urban studies, reviews social media's role in environmental health research focusing on air quality, and identifies methodological gaps. The study analyzes occupant feedback on the X platform (formerly Twitter), exploring how stay-at-home orders during the pandemic's first year affected indoor air quality perceptions, compared to 2019 baseline data. The findings reveal a significant increase in negative perceptions in 2020. This research underscores the importance of social media analytics in shaping environmental health strategies and emphasizes the need to prioritize indoor air quality in design of urban buildings and policy-making. Particularly in the context of global events such as pandemics, this focus is crucial for enhancing overall urban resilience.

Chapter 9 explores occupant-driven urban building energy efficiency, enhanced by ambient intelligence (AmI). It acknowledges the role of human factors—occupancy, behavior, and socioeconomic dimensions—in achieving urban decarbonization goals across a building’s lifecycle, including design, construction, operation, and retrofit. Advances in digital technologies and data-driven approaches, including sensing, computing, and AI, offer new opportunities to understand and reduce energy use in buildings. The chapter reviews energy monitoring methods and systems that encourage positive occupant behaviors, highlighting the benefits of real-time monitoring through AI and IoT-powered AmI systems. These systems aim to support adaptable, user-centered environments and improve occupancy data for future scenario modeling. Emphasizing a holistic approach, it advocates for developing digital twins to supplement models, incorporating occupants in the feedback loop through a socio-technical approach. This spans across building lifecycle stages, encompassing design, planning, policymaking, and operation.

Chapter 10 investigates the complex interplay of social contexts in urban building and transportation energy use. It focuses on how human dynamics, including social determinants of public health (SDPH), mobility, and occupancy, shape urban energy consumption. The research introduces a novel urban energy model using ANNs, augmented with Garson, Lek’s profile, and Partial Dependence Plot (PDP) methods, for an in-depth analysis of energy behaviors across Chicago. Validated through diverse public datasets and cross-validation, this model integrates building and transportation energy use, emphasizing the significance of human dynamics. The findings reveal a pronounced link between SDPH status, mobility, occupancy, and urban energy behavior, identifying household income as a crucial factor. The study underscores the need for a comprehensive evaluation of various urban energy determinants to achieve decarbonization goals in cities, advocating for the use of advanced technologies and detailed analytical approaches to fully understand urban energy use behaviors.

Chapter 11 explores enhancing building performance through virtual reality (VR), particularly in the early design phase. It underscores the importance of initial design decisions in shaping the built environment, individual experiences, and addressing climate change and resource depletion. While metrics for energy consumption and occupant comfort are well-established, qualitative aspects such as aesthetic experience lack established metrics. In light of energy challenges, effective communication and collaboration among stakeholders are vital. The chapter explores VR’s potential and limitations for investigating both quantitative and qualitative design aspects. It reviews existing studies on VR’s application in building performance, spanning visualization, design, education, and research. Additionally, it identifies current limitations and research gaps in VR usage for building performance, aims to support future research and technological advancements, and emphasizes its value in early building design stages.

Chapter 12 delves into the use of digital twins for citywide energy modeling and management, a critical tool for understanding and reducing urban carbon emissions. It explores and conceptualizes urban building energy digital twins (UBEDTs), offering a comprehensive examination of its definition, conceptual framework, architecture, maturity levels, and enabling technologies such as twinning, AI, ML, and interoperability solutions. The study also addresses the limitations of UBEDT development and implementation. By providing an in-depth understanding of UBEDTs, the chapter aims to enhance their adoption in urban

building decarbonization efforts. This investigation not only bridges a significant gap in existing literature but also highlights UBEDTs' potential, especially in utilizing AI-driven techniques. The findings equip researchers and policymakers with insights to foster innovation and progress in urban building energy systems and management.

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